

CY1308 Physics



LECTURE 3

WORK AND ENERGY (Y&F CH 6,7)

Work done by a constant force (SI Unit: 1 J = 1 kg m² s⁻²)

If a force applied to a body or system causes it to have a displacement (in the direction of the force), then we say that it is because the application of the force results in energy input to the body or system.

This energy input is given the formal term, **work done by the force**. It is a **scalar quantity** (i.e. its sign is part of its magnitude).

Work done by a force = force x displacement in the direction of the force.
(Equivalently, component of force in the direction of displacement x displacement)

$$W = \vec{F} \cdot \vec{s}$$

A person pulling a crate along the floor. The work done by the force $\vec{F}$ is $W = Fd \cos \theta$, where $\vec{d}$ is the displacement.

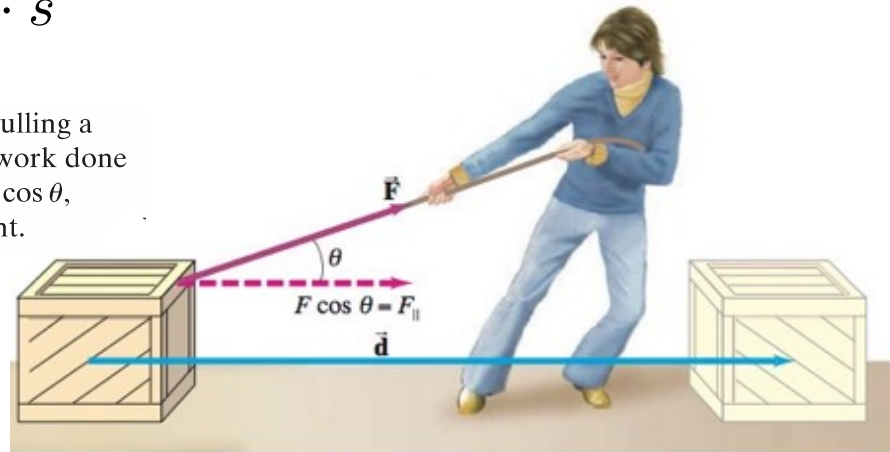


Image source: Giancoli

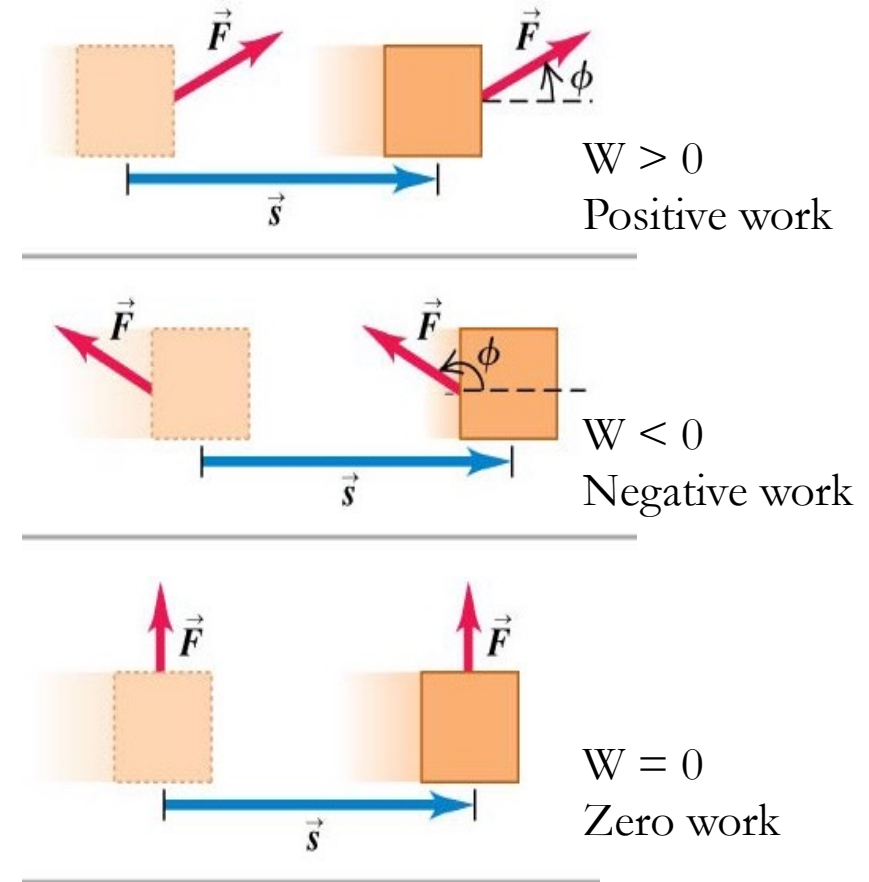


Image source: Young and Freedman

Work done by a variable force in 1D

In the previous slide, work done was defined for a constant force. Consider now, a force applied in 1D.

In the limit of infinitesimally short displacements, the force is constant over the tiny intervals.

Therefore, the work done by the force in the total displacement from x_1 to x_2 is the integral of F_x from x_1 to x_2 :

On a graph of force as a function of position, the total work done by the force is represented by the **area** under the curve between the initial and final positions.

Work done on a particle by a varying x -component of force F_x during straight-line displacement along x -axis

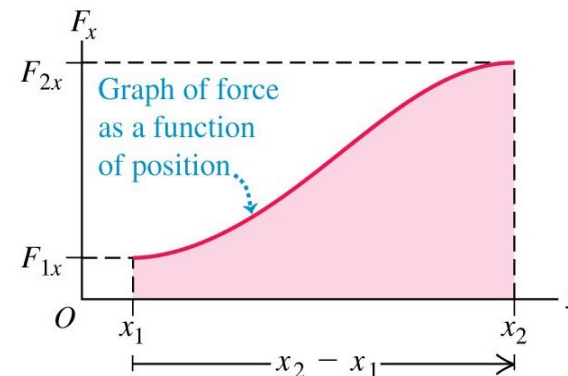
$$W = \int_{x_1}^{x_2} F_x dx$$

Upper limit = final position
Lower limit = initial position
Integral of x -component of force

A particle moves from x_1 to x_2 in response to a changing force in the x -direction.



The force F_x varies with position x ...



... but over a short displacement Δx , the force is essentially constant.

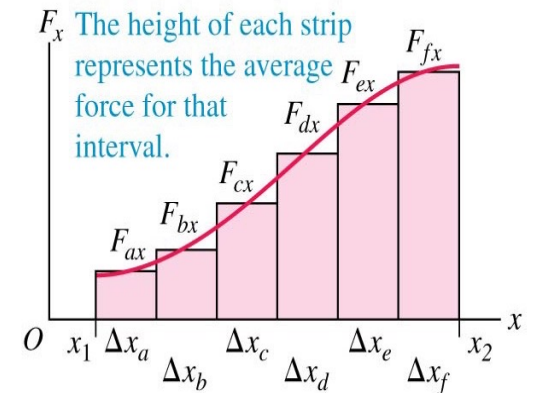


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Work done by a variable force in 2, 3D

Consider now, a variable force applied to a particle, which moves in 2D space from point a to point b.

The path can be divided into small intervals such that in each short interval, the force acting on the particle during each Δl_i is approximately constant.

For each finite interval, the force does an amount of work:

$$\Delta W \approx F_1 \cos \theta_1 \Delta l_1.$$

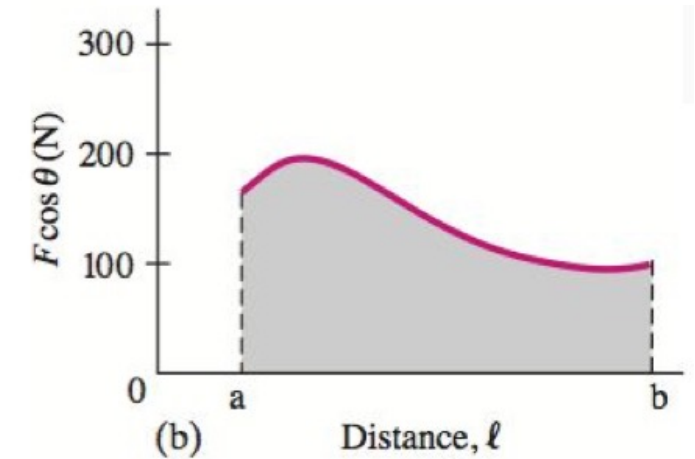
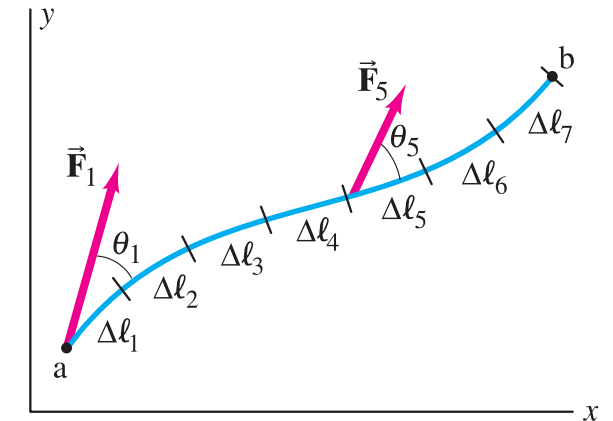
The work done over all the finite intervals from a to b is:

$$W \approx \sum F_i \cos \theta_i \Delta l_i.$$

In the limit of infinitesimally short intervals, the force during each interval is constant. The work done by the force is then given by

$$W = \lim_{\Delta l_i \rightarrow 0} \sum F_i \cos \theta_i \Delta l_i = \int_a^b \vec{F} \cdot d\vec{l}$$

Note: the integral is known as a line integral.



Work done is equal to the area under the curve.

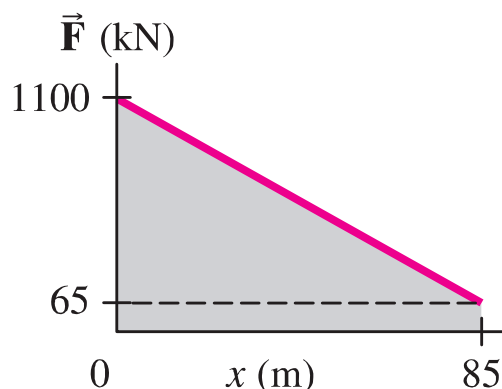
Image source: Giancoli

Example 1: Work done by a force

(II) A 17,000-kg jet takes off from an aircraft carrier via a catapult (Fig. 7–22a). The gases thrust out from the jet's engines exert a constant force of 130 kN on the jet; the force exerted on the jet by the catapult is plotted in Fig. 7–22b. Determine: (a) the work done on the jet by the gases expelled by its engines during launch of the jet; and (b) the work done on the jet by the catapult during launch of the jet.



(a)



(b)

Solution:

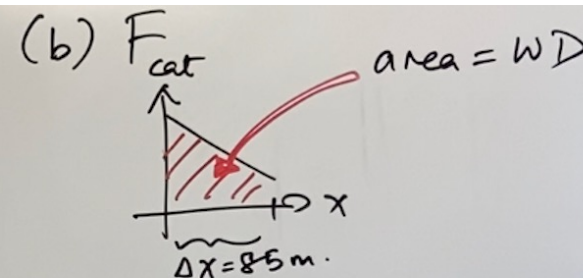
(a) $F_{\text{gas}} = 130 \text{ kN}$

WD by gas

$$= F_{\text{gas}} \Delta x$$

$$= (130 \times 10^3 \text{ N})(85 \text{ m})$$

$$= 11.05 \times 10^6 \text{ J}$$



WD by catapult

$$= \int F_{\text{cat}} dx = (\text{area under curve})$$

$$= \frac{1}{2} (1100 + 65)(85) \text{ kJ}$$

$$= 49.5 \text{ MJ}$$

Note that in the recorded lecture video, the factor of $\frac{1}{2}$ was missing.

Example 2: Work done on a body

Sometimes, instead of work done by a force, because we are interested in the effect of work done the net force on a body, we use the term work done on a body.

Work done on a body = work done by net force on a body

$\sum F = 0$ net work done = 0

(II) A 380-kg piano slides 3.9 m down a 27° incline and is kept from accelerating by a man who is pushing back on it parallel to the incline (Fig. 7-21). Determine: (a) the force exerted by the man, (b) the work done by the man on the piano, (c) the work done by the force of gravity, and (d) the net work done on the piano. Ignore friction.



Solution:

$\theta = 27^\circ$
 $m = 380 \text{ kg}$
 $\Delta x = 3.9 \text{ m}$

$a_{\text{net}} = 0 \Rightarrow F_{\text{net}} = 0$

(d) WD on piano = WD by net force on piano = 0

WD by normal force
 $= N \cos 90^\circ \Delta x$
 $= 0$

WD by man
 $= F_{\text{man}} \cos 180^\circ \Delta x$
 $= -F_{\text{man}} \Delta x$
 $= -mg \sin \theta \Delta x = -6.594 \text{ kJ}$

WD by gravitational force
 $= mg \cdot \Delta \vec{x}$
 $= mg \sin \theta \Delta x$
 $= 6.594 \text{ kJ}$

Source: Giancoli

Work-energy theorem (qualitative)

- The work done by the net force on a particle as it moves is called the **total work** W_{tot} .
- The particle speeds up if $W_{\text{tot}} > 0$, slows down if $W_{\text{tot}} < 0$, and maintains the same speed if $W_{\text{tot}} = 0$.

The **work-energy theorem** states that

The work done by the net force on a particle equals the change in the particle's kinetic energy.

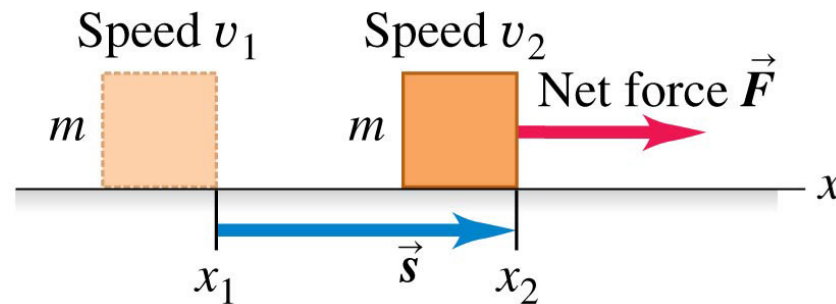


Image source: Young and Freedman

Aside: you will know by now (after years of schooling), that kinetic energy has a pre-factor of $\frac{1}{2}$. Doesn't it seem strange that an “arbitrary” or “nice” constant is in front of the factors m and v^2 ? Where does it come from? Or was it defined this way?

Work-energy theorem (derivation)

Consider a net force $\mathbf{F}$ on a particle moving from point a to point b.

The net work done by the force is given by

$$W_{\text{net}} = \int \vec{\mathbf{F}}_{\text{net}} \cdot d\vec{\ell} = \int F_{\text{net}} \cos \theta \, d\ell = \int F_{\parallel} \, d\ell,$$

where the subscript $\parallel$ represents the component of force parallel to each infinitesimally small displacement. Since

$$F_{\parallel} = ma_{\parallel} = m \frac{dv_{\parallel}}{dt} = m \frac{dv}{dt}$$

where subscript $\parallel$ represents the component tangent to the curve.

Because the particle is moving along the curve, the velocity has only a component tangent to the curve, $\vec{v} = \vec{v}_{\parallel}$, hence $dv_{\parallel}/dt = dv/dt$.

Therefore, the work done by $\mathbf{F}$ is:

$$\begin{aligned} W_{\text{net}} &= \int_i^f F_{\parallel} dl = \int_i^f m \frac{dv}{dt} dl = \int_i^f m \frac{dv}{dl} \frac{dl}{dt} dl && \text{(using chain rule)} \\ &= \int_i^f m v \frac{dv}{dl} dl = \int_i^f m v dv = \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2 \end{aligned}$$

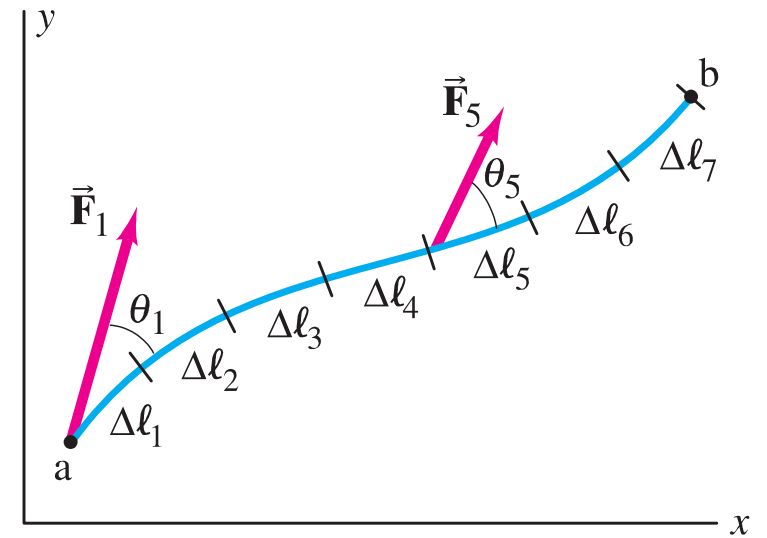


Image source: Giancoli

Example 3: Application of work-energy theorem

(II) A mass m is attached to a spring which is held stretched a distance x by a force F (Fig. 7-28), and then released. The spring compresses, pulling the mass. Assuming there is no friction, determine the speed of the mass m when the spring returns: (a) to its normal length ($x = 0$); (b) to half its original extension ($x/2$).

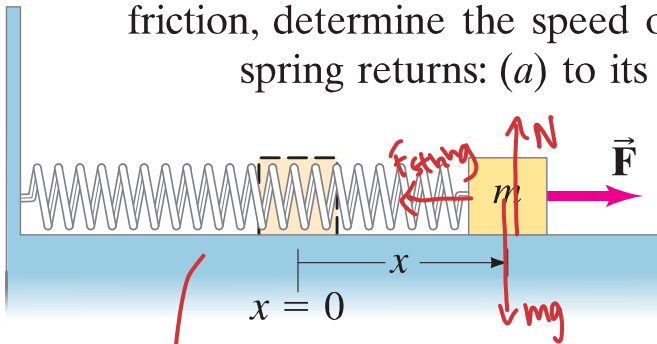


FIGURE 7-28
Problem 57.

Smooth.

$F_{\text{spring}} = kx$
spring constant

Solution:

Ex 3.

$\Delta KE + \Delta PE = 0$

(a) $x_f = 0; v_0 = 0$	(b) $x_f = \frac{1}{2}x_0$ $v_0 = 0$
$v_f = \sqrt{\frac{k}{m}} x_0$	$v_f = \sqrt{\frac{3k}{4m}} x_0$

Apply Work-Energy theorem.

WD by spring force = ΔKE

$$\int_{x_0}^{x_f} \vec{F}_{\text{spring}} \cdot d\vec{x} = \frac{1}{2} m (v_f^2 - v_0^2)$$

$$-\int_{x_0}^{x_f} kx \cdot \hat{x} dx = \frac{1}{2} m (v_f^2 - v_0^2)$$

$$-\int_{x_0}^{x_f} kx dx =$$

$$\frac{1}{2} k (x_0^2 - x_f^2) = \frac{1}{2} m (v_f^2 - v_0^2)$$

Change in KE
Change in elastic PE

Conservative and non-conservative forces

We will find it important to categorize forces into two types: conservative and non-conservative forces. By definition a force is **conservative** if

the work done by the force on an object moving from one point to another depends only on the initial and final positions of the object, and is independent of the particular path taken.

Equivalently, a force is conservative if the net work done around any closed path is zero.

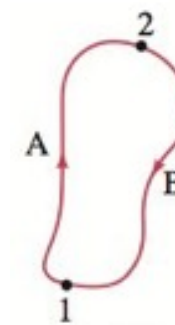
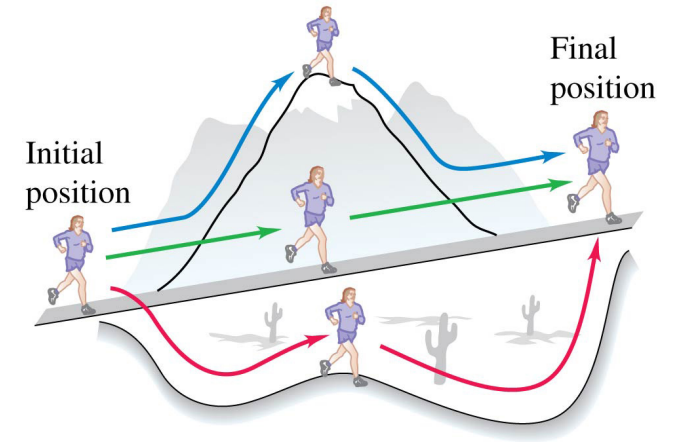
The work done between two points by any conservative force

- a) can be expressed in terms of a **potential energy function**.
- b) is reversible.
- c) is independent of the path between the two points.
- d) is zero if the starting and ending points are the same.

A force (such as friction) that is not conservative is called a **nonconservative force**, or a **dissipative force**. Gravitational force and spring force are conservative.

Image source: Young and Freedman

Because the gravitational force is conservative, the work it does is the same for all three paths.



A closed path. The object goes out from point 1 via A to 2 and back to point 1 via B.

The work done by a conservative force around a closed path is zero.

Gravitational potential energy and work done

- Consider a system comprising the Earth and an object of mass m . The object is lifted vertically by a displacement $\vec{s}$.
- In order for this to happen, an external force needs to be applied (by an “agent” external, i.e. not belonging, to the system of Earth and object).
- This external force needs to be equal and opposite to the weight of the object: $\vec{F}_{\text{ext}} = -m\vec{g}$
(Of course, to start it moving, i.e. to accelerate it, from rest, at the initial application the external force must be *just* larger than the weight.)
- The work done by the external force is given by:

$$\begin{aligned} W_{\text{ext}} &= \int_{y_1}^{y_2} \vec{F}_{\text{ext}} \cdot d\vec{y} = \int_{y_1}^{y_2} mg\hat{y} \cdot \hat{y}dy \\ &= \int_{y_1}^{y_2} mgdy \quad \leftarrow \text{Note: in the recorded video, this intermediate step had a sign error.} \\ &= mg(y_2 - y_1) \end{aligned}$$

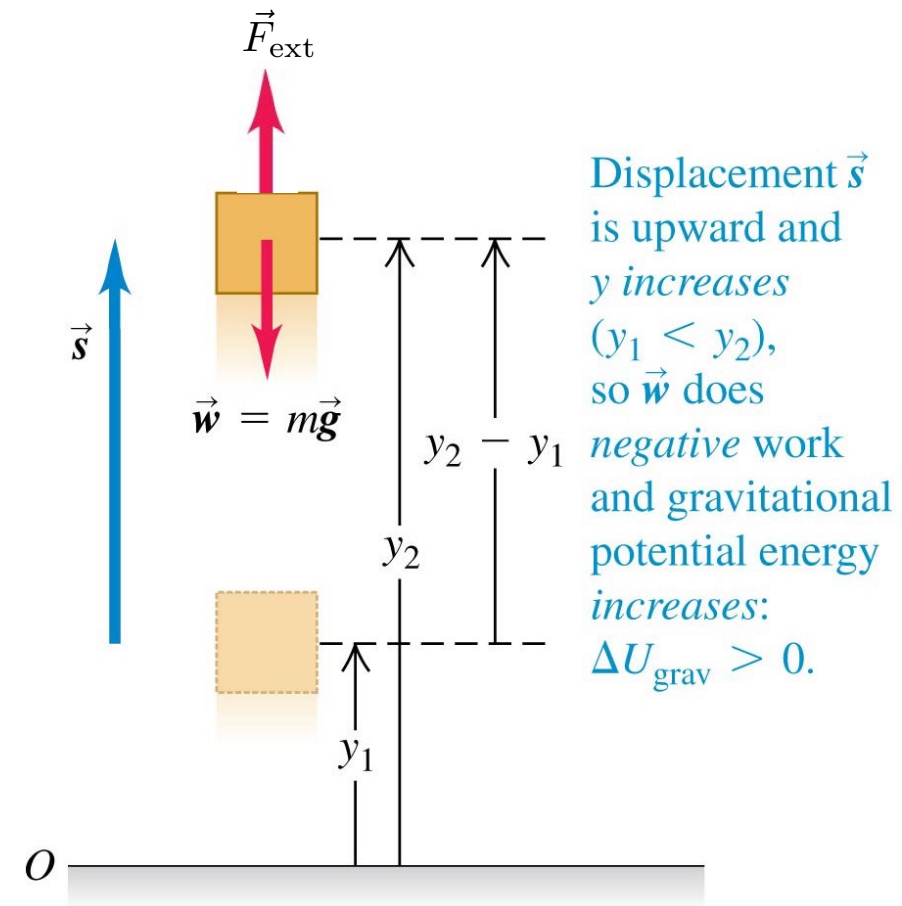


Image source: Young and Freedman

Gravitational potential energy and work done

- Once at this new height, this work or energy input can be recovered, and the object has the ability to do work. We can say that the work done in lifting the object has been stored as gravitational potential energy. Therefore, we define

the **change in gravitational potential energy** of an object is the work done by an external force in moving the object,

$$\Delta U = W_{\text{ext}} = \int \vec{F}_{\text{ext}} \cdot d\vec{l}$$

Or equivalently,

the **change in gravitational potential energy** of an object is the negative of the work done by the gravitational force.

$$\Delta U = -W_{\text{grav}} = - \int \vec{F}_{\text{grav}} \cdot d\vec{l}$$

- Note that it is only the change in potential energy that can be defined, since two end positions need to be referenced in the integral. This means that we are free to define a reference position for the potential energy. We may take the Earth's surface to be zero gravitational potential energy.

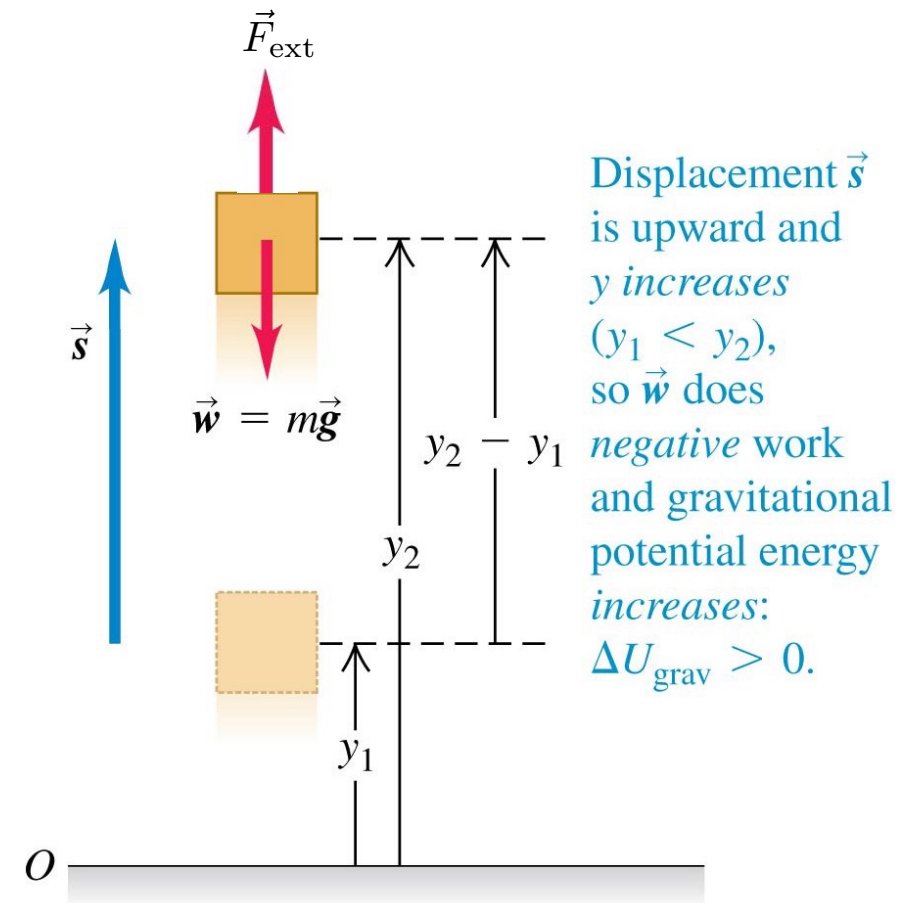


Image source: Young and Freedman

Potential energy and work done

- To generalize the previous derivation which considered 1D motion, we can consider an arbitrary path.
- Examining each infinitesimal displacement, we note that only the vertical component of displacement goes into the calculation of work, since the force (external or gravitational) is vertical.
- Therefore, the definition of potential energy from the previous slide still holds.
- Also, we can generalize from gravitational force to any conservative force F :

$$\Delta U = - \int \vec{F} \cdot d\vec{l}$$

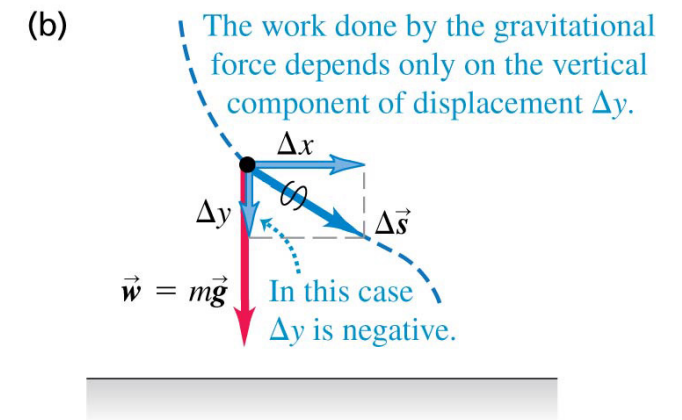
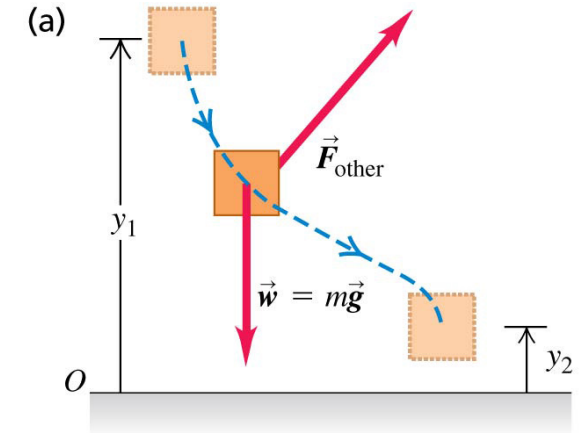
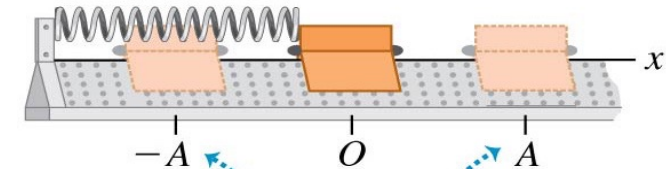


Image source: Young and Freedman

Mechanical energy

- Mechanical energy is defined as the sum of kinetic (K) and potential (U) energy.
- If only conservative forces are doing work, the total mechanical energy of a system is constant (it is conserved).
- In contrast, total energy is always conserved.



The limits of the glider's motion are at $x = A$ and $x = -A$.

On the graph, the limits of motion are the points where the U curve intersects the horizontal line representing total mechanical energy E .

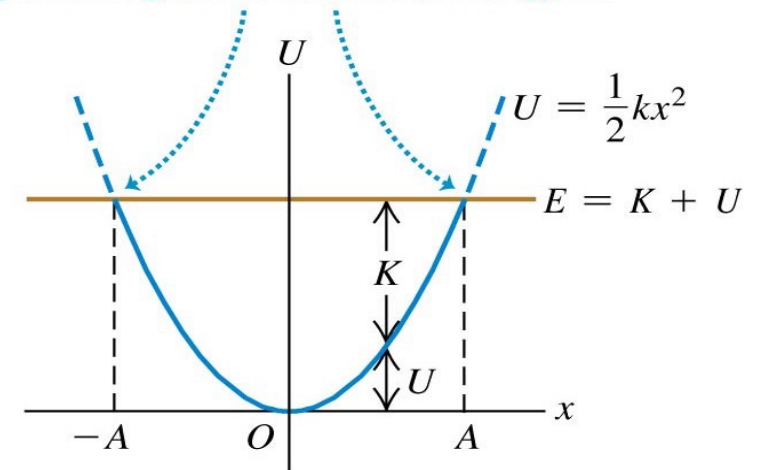


Image source: Young and Freedman

Principle of conservation of energy

Nonconservative forces do not store potential energy, but they do change the **internal energy** of a system.

The law of conservation of energy means that energy is never created or destroyed; it only changes form.

This law can be expressed as

$$\Delta K + \Delta U + \Delta U_{\text{int}} = 0.$$

Or equivalently

$$\Delta K + \Delta U + (\text{change in all other forms of energy}) = 0$$

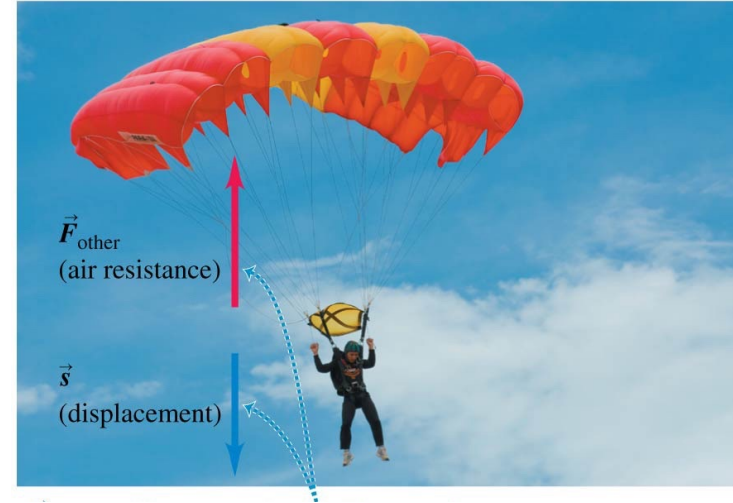


Image source: Young and Freedman

Sometimes it is hard to define internal energy as in this case.

We can always use the work energy theorem:

$$W_{\text{grav}} + W_{\text{air}} = \Delta K$$

$$-\Delta U + W_{\text{air}} = \Delta K$$

$$W_{\text{air}} = \Delta K + \Delta U$$

i.e. change in mechanical energy (RHS) = WD by air resistance (LHS), which we know is negative given the opposite direction of air resistance to displacement.

Trust in Energy Conservation



Trust in Physics

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Walter Lewin: <https://www.youtube.com/watch?v=xXXF2C-vrQE>

Elastic potential energy

Spring force is given by Hooke's Law (within the proportionate limit) as: $\vec{F}_{\text{spring}} = -k\vec{x}$,
 where k is the spring constant (the bigger k , the stiffer the spring)
 and $\vec{x}$ is the extension from the unstretched length.

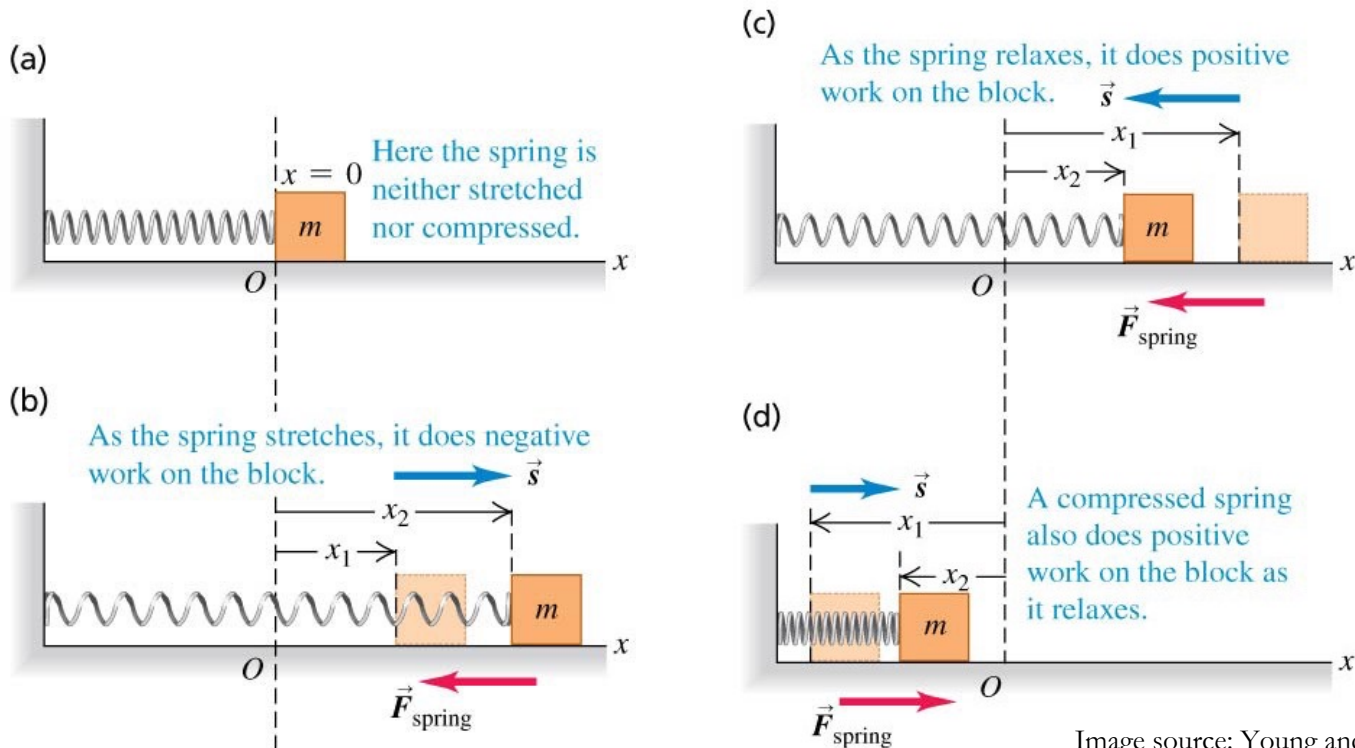
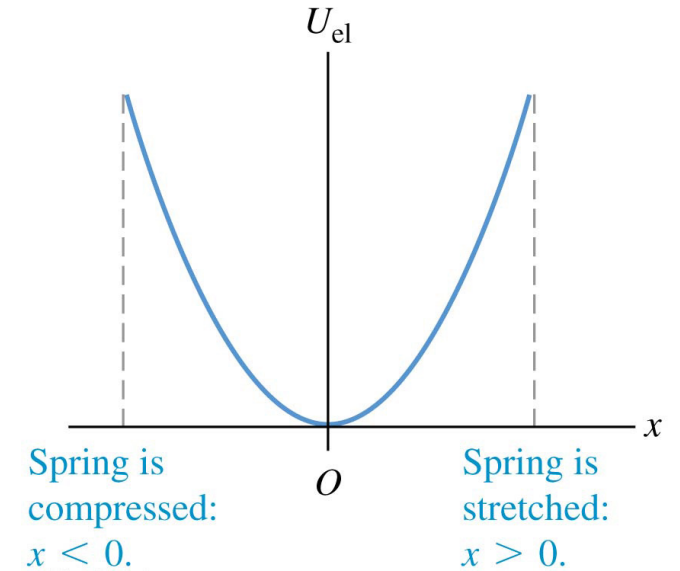


Image source: Young and Freedman



The elastic potential energy can be derived:

$$\begin{aligned}
 U_{\text{el}} &= - \int_0^x \vec{F}_{\text{spring}} \cdot d\vec{x} \\
 &= \int_0^x kx \, dx \quad (\because d\vec{x} \equiv \hat{x} \, dx) \\
 &= \frac{1}{2} kx^2
 \end{aligned}$$

Force and potential energy (1D): differential form

In 1D, a conservative force can be written in terms of its potential energy function in differential form as:

Force from potential energy:

In one-dimensional motion, the value of a conservative force at point x ...

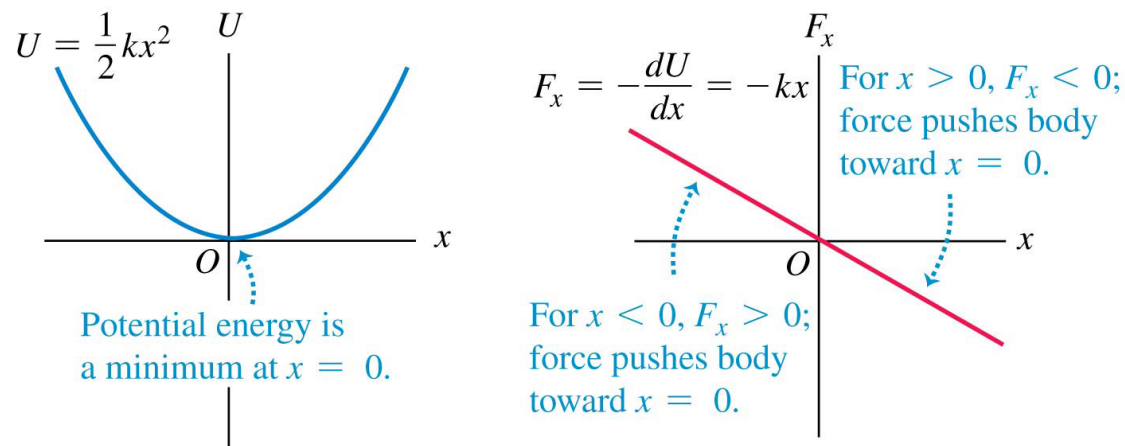
$$F_x(x) = -\frac{dU(x)}{dx}$$

... is the negative of the derivative at x of the associated potential-energy function.

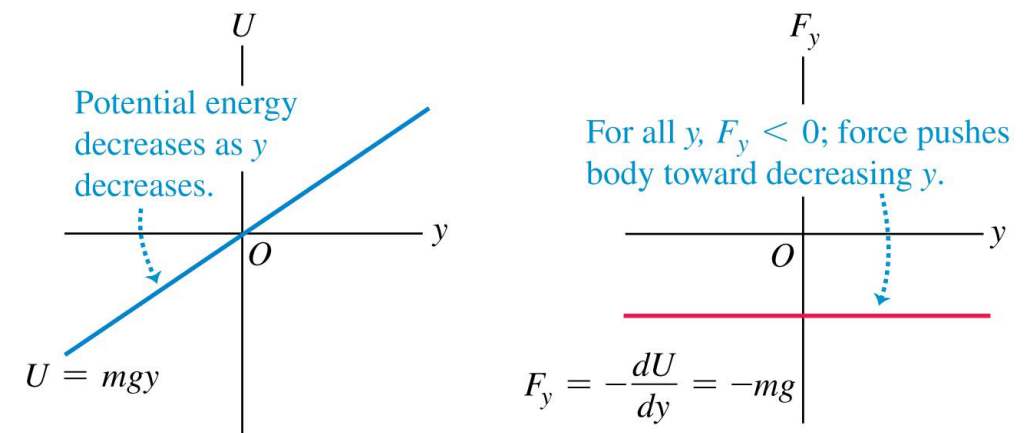
- In regions where $U(x)$ changes most rapidly with x , this corresponds to a large force magnitude.
- When $F_x(x)$ is in the positive x -direction, $U(x)$ decreases with increasing x .
- A conservative force always acts to push the system toward lower potential energy.

Image source: Young and Freedman

(a) Elastic potential energy and force as functions of x



(b) Gravitational potential energy and force as functions of y



Force and potential energy (3D): differential form

In 3D, the components of a conservative force is related to its potential energy function with partial derivatives:

Force from potential energy: In three-dimensional motion, the value at a given point of each component of a conservative force ...

$$F_x = -\frac{\partial U}{\partial x} \quad F_y = -\frac{\partial U}{\partial y} \quad F_z = -\frac{\partial U}{\partial z}$$

... is the negative of the partial derivative at that point of the associated potential-energy function.

When we take the partial derivative of U with respect to each coordinate, multiply by the corresponding unit vector, and then take the vector sum, this is called the **gradient** of U (**which is a vector quantity**)

Force from potential energy: The vector value of a conservative force at a given point ...

$$\vec{F} = -\left(\frac{\partial U}{\partial x}\hat{i} + \frac{\partial U}{\partial y}\hat{j} + \frac{\partial U}{\partial z}\hat{k}\right) = -\vec{\nabla}U$$

... is the negative of the gradient at that point of the associated potential-energy function.

Force and potential energy relationship (3D)

In the previous slide, we used the “del” operator, which is defined in Cartesian coordinates as:

$$\nabla \equiv \hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z}$$

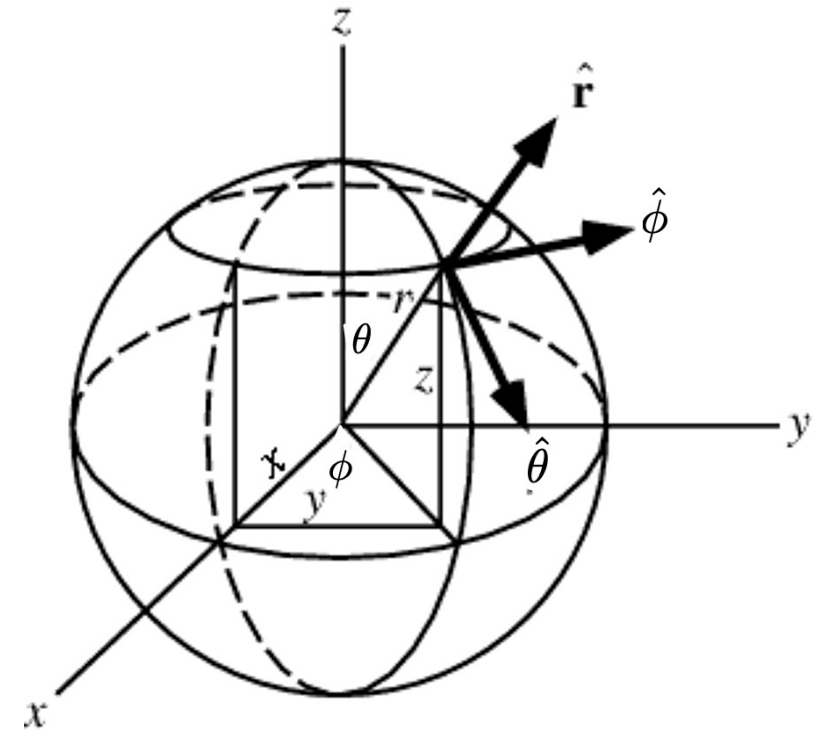
The del operator needs a function on the right to operate on. By itself, it can't do anything, so it is meaningless to write it on its own (except as a definition).

In a curvilinear coordinate system, such as the spherical coordinate system (right), the del operator can be derived via a coordinate transformation to be:

$$\nabla \equiv \hat{r} \frac{\partial}{\partial r} + \hat{\theta} \frac{1}{r} \frac{\partial}{\partial \theta} + \hat{\phi} \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \quad (\text{spherical coords})$$

This is useful for systems with spherical symmetry, such as gravitational forces on a planetary scale.

Note: You do not need to memorise the expression for del.



A point in 3D in the spherical coordinate system is specified by (r, θ, ϕ) . The unit vectors $(\hat{r}, \hat{\theta}, \hat{\phi})$ move with the point.

Power (SI Unit: $1 \text{ W} = 1 \text{ J s}^{-1} = 1 \text{ kg m}^2 \text{ s}^{-3}$)

Power is the *rate* at which work is done.

Average power is:

Average power during time interval Δt $\rightarrow P_{\text{av}} = \frac{\Delta W}{\Delta t}$

ΔW Work done during time interval
 Δt Duration of time interval

Instantaneous power is:

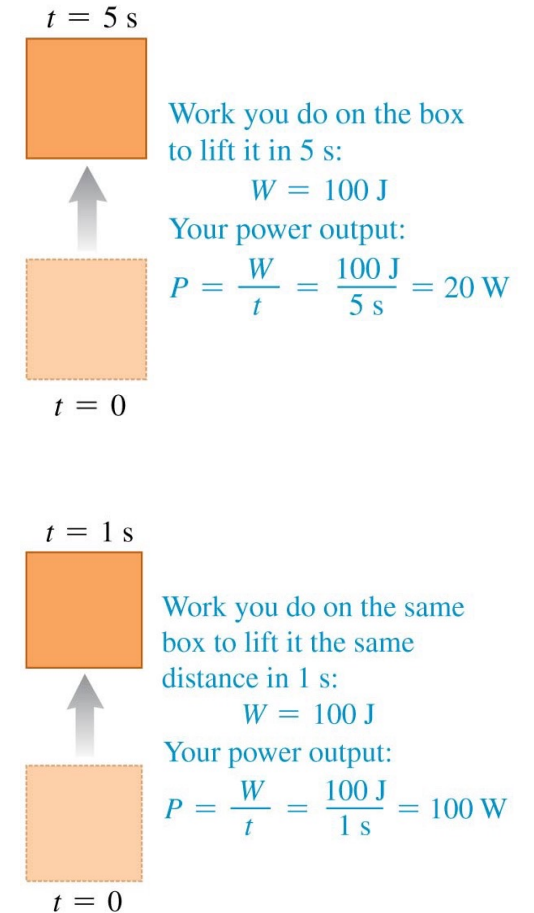
Instantaneous power $\rightarrow P = \lim_{\Delta t \rightarrow 0} \frac{\Delta W}{\Delta t} = \frac{dW}{dt}$

Time rate of doing work
Average power over infinitesimally short time interval

Mechanical power is:

Instantaneous power for a force doing work on a particle $\rightarrow P = \vec{F} \cdot \vec{v}$

$\vec{F}$ Force that acts on particle
 $\vec{v}$ Velocity of particle



Check your intuition: how high does it rise?

A pendulum bob on the right, released at the level indicated will have its string stopped by the bar indicated by the arrow.

How high will the pendulum bob rise to on the left side?

